

NGUYEN KHAC NAM

FRONTEND DEVELOPER

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SUMMARY

Frontend Developer with 3+ years of experience building enterprise-scale web applications using React and TypeScript. Experienced in leading frontend development for business-critical modules, designing scalable architectures, building reusable component systems, and optimizing application performance. Passionate about creating maintainable software, driving engineering excellence, and collaborating across teams to deliver reliable solutions that solve complex business challenges.

EDUCATION

Hanoi University of Mining and Geology 2019 – 2023
Engineer's Degree in Information Technology

SKILLS

Languages: JavaScript, TypeScript, Java

Frontend

- **Frameworks/Libraries:** React, Next.js, Vue.js
- **State Management:** Redux, MobX, SWR, React Query
- **Form & Validation:** React Hook Form, Formik, Yup
- **UI Libraries:** TailwindCSS, Material UI, Ant Design
- **Realtime:** WebSocket, Server-Sent Events
- **Charts:** ECharts, Chart.js, Recharts
- **Performance Optimization:** Code Splitting, Lazy Loading, Bundle Optimization, Rendering Optimization
- **Architecture & System Design:** Scalable Frontend Architecture, Component-Driven Design, Reusable UI Components, Feature-Based Module Organization
- **Testing:** Jest, React Testing Library
- **Build Tools:** Webpack, RSBUILD, Vite
- **Monorepo:** Nx, Turborepo, pnpm Workspace

Backend & Integration

- **Frameworks/Platform:** Node.js, Express, NestJS, Spring Boot
- **Database:** MySQL, PostgreSQL, MongoDB

EXPERIENCE

Frontend Developer (Onsite) Jan 2026 - Present
Viettel High Tech

Project: Vtracking Viettel

Overview: VTracking is a vehicle tracking and online monitoring solution that utilizes GPS technology, mobile data transmission, and Geographic Information Systems (GIS) to provide real-time vehicle tracking and location monitoring on digital maps.

Team size: 20

Tech Stack:

- **Frontend:** jQuery, Bootstrap, Twirl Template (SSR), Viettel Maps API, WebSocket, Video Streaming (JSMpeg, Jessibuca)
- **Backend:** Play Framework, Microservices (Go, PostgreSQL, MongoDB, Kafka, ClickHouse)

Responsibilities & Achievements:

- Owned the frontend implementation of live camera monitoring and playback features, including video rendering, stream controls, player interactions, and error handling for various streaming states.
- Designed and maintained interactive monitoring interfaces integrating Viettel Maps API, real-time vehicle tracking, camera visualization, and operational management workflows.
- Integrated WebSocket-based communication to deliver real-time updates for vehicle locations, camera statuses, alerts, and monitoring events.
- Implemented advanced map-based features including vehicle tracking, route visualization, geofencing, marker clustering, and dynamic layer management.
- Collaborated with backend and platform teams to define streaming APIs, WebSocket contracts, camera management workflows, and real-time monitoring requirements.

Frontend Developer

Jul 2024 - Jan 2026

VTI Joint Stock Company

Project: Manufacturing Execution System (MES-X)

Overview: A comprehensive management solution that provides an overview of the entire manufacturing process with 5 modules: MESCore, WMS-X, MMS-X, QMS-X, PMS-X

Team size: 60

Tech Stack:

- **Frontend:** React, Redux, Redux-Saga, SWR, React Hook Form, Formik, Yup, Material UI, Recharts, Lodash, Date-fns, React Virtuoso, React Window
- **Backend:** Microservices (NestJS, PostgreSQL, MongoDB, Kafka, Redis, ClickHouse)

Responsibilities & Achievements:

- Led frontend development for MESCORE and WMS-X modules, managing a team of 5 frontend engineers and coordinating delivery within a 15-member frontend organization.
- Owned sprint planning, effort estimation, task assignment, code reviews, and pull request governance to ensure on-time and high-quality releases.
- Designed and maintained scalable frontend architecture for MESCORE and WMS-X, including component structure, routing, state management, design system integration, and shared utilities.
- Established standardized data management patterns across Global State (Redux), Server State (SWR), and Local State to improve maintainability, scalability, and developer productivity.
- Built a centralized API communication layer with request abstraction, interceptor pipelines, authentication handling, and unified error management for microservice-based systems.
- Developed and maintained 20+ reusable UI components, shared hooks, and business widgets used across six enterprise platforms: MESCORE, WMS-X, QMS-X, MMS-X, PMS-X, and TMS-X.
- Created reusable form and data-table frameworks using React, Formik, React Hook Form, and React Virtuoso to standardize complex business workflows across multiple products.
- Implemented core manufacturing workflows, including Production Orders, Manufacturing Orders, Production Programs, Bill of Materials (BOM), Bill of Operations (BOO), Routing, and Master Planning processes.
- Developed warehouse management workflows covering inventory requests, goods receipt and issue operations, warehouse transfers, put-away, picking, stocktaking, and inventory adjustment processes.
- Optimized application performance through virtualization, memoization, lazy loading, and rendering optimization techniques, enabling smooth user experiences with datasets exceeding 300,000 records.

- Built operational dashboards, KPI reporting systems, and production analytics modules to support real-time monitoring and business decision-making.
- Engineered highly dynamic form-table solutions supporting 40+ interdependent fields, advanced validation logic, and high-volume manufacturing data entry operations.
- Integrated Server-Sent Events (SSE) and Socket.IO to deliver real-time features such as OCR processing notifications, workflow status tracking, and live production dashboard updates.
- Collaborated with backend engineers, business analysts, QA teams, and product stakeholders to define API contracts, align microservice integrations, and ensure reliable end-to-end delivery.
- Enforced engineering best practices through coding standards, Git Flow adoption, code reviews, technical mentoring, knowledge sharing, and continuous improvement initiatives.

Fullstack Developer (Onsite)

Oct 2022 - May 2024

Vietnam Maritime Corporation (VIMC)

Project: VIMC E-Office

Overview: VIMC E-Office is the internal information system of the Vietnam Maritime Corporation (VIMC). It provides features for document processing, task management, records management, and online internal communication

Team size: 15

Tech Stack:

- **Frontend:** ReactJS, MobX React, Ant Design, Styled-Components, Lodash, Moment.js
- **Backend:** Java(Spring Boot), MySQL

Responsibilities & Achievements:

- Developed new pages and features for document workflows and approval systems using React, Ant Design, and MobX
- Extended and maintained existing CKEditor 5 integrations, adding new functionalities and resolving bugs in document editing modules
- Built and updated reusable UI components with Styled-Components to ensure design consistency across new features
- Integrated RESTful APIs with backend services for document processing, record management, and task tracking
- Implemented Excel import/export features using ExcelJS to support document reporting and data operations
- Collaborated with cross-functional team using Git, participating in code reviews and feature development cycles
- Developed new backend APIs with Spring Boot and MySQL to support document and task management modules
- Fixed bugs and extended existing backend features while gaining hands-on experience in Java and Spring Boot development
- Implemented file upload/download services and data validation logic for document management

Project: VIMC LINES

Oct 2022 - May 2024

Overview: VIMC LINES provides container transportation services, port operations, and logistics supply chain services. VIMC LINES aims to develop an integrated supply chain of sea transport - port operations - logistics services.

Team size: 12

Tech Stack:

- **Frontend:** React, MobX, Ant Design, Styled-Components, Lodash, Moment.js, React Query, Axios, i18next, Tiptap Editor, ExcelJS, Styled Icons
- **Backend:** Node.js (Express), Prisma ORM, PostgreSQL / MongoDB, Axios, Redis (ioredis), Nodemailer, Puppeteer, JWT, EJS, ExcelJS, Docx, Day.js, Lodash

Responsibilities & Achievements:

- Contributed to developing new pages and features using React, MobX, and Ant Design for logistics and port operation modules
- Assisted in building reusable UI components with Styled-Components and Styled Icons to maintain consistent interfaces
- Integrated RESTful APIs with Axios and React Query for container tracking and logistics data
- Supported feature updates for Tiptap Editor used in document editing and internal communication
- Implemented Excel import/export and multilingual features with ExcelJS and i18next under team guidance
- Contributed to developing API endpoints using Express and Prisma ORM for booking and logistics features
- Assisted in maintaining PostgreSQL database schemas for vessels, containers, and shipment data
- Helped implement file generation features using Puppeteer, Docx, and ExcelJS for shipping documents
- Collaborated with team members to define API contracts and resolve integration issues between frontend and backend modules

Intern Next.js Developer

Aug 2022 - Sep 2022

Thinkdiff Artificial Intelligence Company Limited

Project: SamNotes

Overview: A web app to note information, ideas, tasks and access the application anytime, anywhere.

Team size: 7

Tech Stack:

- **Frontend:** Next.js 13, TailwindCSS, Ant Design
- **Backend:** Python(Flask), MySQL

Responsibilities & Achievements:

- Developed note management features (create, read, update, delete) using Next.js 13 (TypeScript) and Ant Design.
- Implemented App Router, dynamic routing, and integrated frontend components with REST APIs for note operations.
- Built responsive user interfaces using TailwindCSS, ensuring consistent user experience across devices.
- Applied Next.js best practices including layouts, metadata management, Server Components, and Client Components.
- Collaborated in an Agile team environment, participating in daily meetings, Git workflow, code reviews, and cross-functional discussions.